INCH-POUND

MIL-C-44424(GL)
1 August 1991

MILITARY SPECIFICATION

CLOTH, CHEMICAL PROTECTIVE

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of cloth treated for chemical protection.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 5035 - Breaking Force and Elongation of Textile Fabrics (Strip Force)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material.

3.2.1 Base cloth. The base cloth shall be a material consisting of regular or high tenacity aramid fibrids and polyester fibers properly constructed so that when finshed meets the requirements specified in table I.

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3.2.2 Finish. The base cloth shall be given an approved finish or coating (see 6.4). The finish or coating shall be formulated so that when applied to the cloth specified in 3.2.1, the finished cloth meets the requirements specified in 3.3.

3.2.3 Thread. The finished cloth shall be stitched with an elastomeric thread in such a manner as to meet the initial stretch requirements specified in 3.3.

3.3 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Physical requirements

Characteristic	Requirement	
	Minimum	Maximum
Weight, oz./yd.^{2}	14.0	17.0
Thickness (mils)		
@ 0 percent stretch	100	120
Air permeability, ft.^{3}/min./ft.^{2}	20	26
Burst strength, (lbs)	54.0	-
Initial stretch, percent:		
Machine direction	50.0	-
Cross machine direction	25.0	-
Abrasion resistance, (cycles)		
Face side	1000	-
Back side	770	-
Carbon loading, (g/m^2)		
@ 0 percent stretch	160.0	-
Dynamic CCl4	1.8	-

3.3.1 Width. The width of the relaxed finished cloth shall be 44 ± 1 inches, between pin or clip marks, when tested as specified in 4.4.3.

3.4 Face identification. The side of the cloth opposite the rib effect side shall be considered the face side.

3.5 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in lengths of 80 ± 1 yards, with no piece less than 20 yards. The cloth shall be put-up in rolls in accordance with 5.1.

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3.6 Workmanship. The finished cloth shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2 and tested for the characteristics specified in 4.4.3. The presence of any defect or failure of any test shall be cause for rejection of the first article. The contractor shall be responsible for the necessary corrections to the manufacturing process prior to resubmittal to the Government.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard visual examination. The material shall be examined for the defects listed in table II. The lot size shall be expressed in units of linear yards of material. The sample unit shall be one 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects. The required yardage shall be examined on the face side. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof. All defects shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample yardage shall be apportioned equally among the selected rolls.

TABLE II. End item visual defects

Defect	Classification Major	Minor
Hole, cut, or tear	101	
Lump or imbedded matter greater than 1/2 inch in diameter		201
Permanent or set-in crease or wrinkle		202
Material spliced or lapped	102	
Excessive coating, coating bands or stop marks across width	103	
Edges not straight or uniform:		
Edges rolled, rippled, scalloped, nicked or cut, or dimensional distortation	104	
Any spot, stain, or streak more than 1 inch long		203
Material puckered or finish not uniform		204
Abrasion mark which extends partially through material, or thin place	105	
Missing elastomeric thread:		
1/4 inch to 1/2 inch in any direction		205
greater than 1/2 inch in any direction	106	

4.4.2.2 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the roll ticket shall be considered a length defect. The lot shall be

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rejected if two or more rolls in the sample are defective with respect to length or if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on the sample unit for test purposes, as specified in the applicable test method. The sample unit shall be 3 yards full width. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
12,000 or less	2
12,001 up to and including 20,000	3
20,001 and over	5

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Weight	3.3	5041
Thickness	3.3	5030
Air permeability	3.3	5450
Burst strength	3.3	5120
Initial Stretch		
Machine direction	3.3	ASTM D 5035 1/
Cross machine direction	3.3	ASTM D 5035 1/
Abrasion resistance		
Face side	3.3	5306 2/
Back side	3.3	5306 2/
Carbon loading	3.3	3/
Dynamic CCl4	3.3	4.5.1
Width	3.3.1	4/

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1/ The tester shall be operated at a uniform pulling speed of 2 ± 0.5 in/min.

2/ Test for destruction.

3/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4/ Measurements shall be taken on relaxed material at least 1 yard from end of roll and reported to the nearest 1/8 inch.

4.4.4 Packaging inspection. The sampling and inspection of the preservation, packaging, and marking shall be in accordance with the requirements of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Carbon tetrachloride adsorption.

4.5.1.1 Apparatus and reagents.

4.5.1.1.1 Vapor generator. Any apparatus can be used that generates a carbon tetrachloride/nitrogen gas mixture which contains 4 to 6mg of carbon tetrachloride per minute. Once established, the rate must be maintained at ± 0.5 mg/minute (see 6.6).

4.5.1.1.2 Diluent. Prepurified nitrogen shall be used as the diluent gas for the concentrated carbon tetrachloride vapor generated. The total flow rate of 0.9 to 1.1 liter/minute through each test cup must be maintained at ± 0.03 liter/minute. The rate of flow can be controlled by use of a critical orifice or other suitable flow controlling device.

4.5.1.1.3 Constant temperature cabinet. The constant temperature cabinet shall be large enough to hold the system, as shown on figure 1, and shall be capable of maintaining a temperature of 32° ± 1 $^{\circ}$C at the distribution manifold and 32° ± 2 $^{\circ}$C at the sample cups.

4.5.1.1.4 Fabric sample cup. The fabric sample cup shall conform to the requirements shown on figure 2.

4.5.1.1.5 Flat elastomer ring. Each sample holder cup requires three rings to prevent seal leakage. Rings shall be cut from neoprene or butyl rubber sheet of 30 mil thickness with 40-50 Shore "A" hardness. Prepare two rings 13.80 cm OD X 11.28 cm ID and one ring 13.80 cm OD X 12.70 cm ID for each sample cup.

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4.5.1.1.6 Pyrolyzer. The platinum pyrolyser shall conform to the requirements shown on figure 3. The pyrolyzer shall be used between the outlet of the sample cup and the inlet of the "break point" indicator bubbler. It shall be heated by applying 10 volts AC across the terminals during the test.

4.5.1.1.7 Bubbler. The bubbler shall conform to the requirements shown on figure 4.

4.5.1.1.8 Schwartz-type drying tube. The tube shall be filled with activated charcoal as specified in 4.5.1.1.9, loosely packed as shown on figure 5. Glass wool, loosely packed, shall be placed on top of the activated carbon.

4.5.1.1.9 Charcoal, coconut, activated, 6 to 14 mesh.

4.5.1.1.10 Potassium iodide solution. Dissolve 2.5 grams of potassium iodide in a liter of deionized water. Store in an amber bottle.

4.5.1.1.11 Starch solution. A commercial stabilized starch indicator solution may be purchased, otherwise, prepare a starch solution by dissolving 5 grams of soluable starch in 50 mL of boiling deionized water while stirring. Add the 50 mL of concentrated starch solution to 450 mL of boiling deionized water. Cool and store in a refrigerator. A fresh solution must be prepared whenever cloudiness or mold growth develops.

4.5.1.1.12 Starch/iodide reagent solution. Add 9 mL of potassium iodide solution (4.5.1.1.10) and 1 mL of starch solution (4.5.1.1.11) to each bubbler tube, as needed, during the testing period.

4.5.1.1.13 Analytical balance.

4.5.1.1.14 "C" type clamp.

4.5.1.1.15 Ice bath.

4.5.1.2 Determination of carbon tetrachloride concentration.

4.5.1.2.1 Tare weight of drying tube. The schwartz-type drying tube, filled with charcoal and prepared as shown on figure 5, shall be conditioned for 10 minutes or more using the apparatus with a flow of 5 $\pm$ 1 mg/minute carbon tetrachloride vapor in nitrogen. The drying tube replaces the sample cup for this procedure. Remove the drying tube from the apparatus and place near the weighing balance. After 10 minutes weigh the drying tube. All newly filled drying tubes must be conditioned. The last weight of the drying tube should be used as the initial weight for the next determination. NOTE: The carbon tetrachloride test vapor must always enter the drying tube through the same stopcock.

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4.5.1.2.2 Calibration of carbon tetrachloride concentration. Pass the diluted carbon tetrachloride vapor from the generator manifold at a steady rate between 0.9 to 1.1 liter/minute through a tared Schwartz-type drying tube at each sample port. At 19.7 minutes stop the vacuum pump. At 20 minutes close the stopcocks and remove the Schwartz tubes to cool for 10 minutes at ambient temperature. Open and close one stopcock to equalize pressure and weigh to ± 0.1 mg. Calibration conducted at the beginning and end of the day shall be averaged for the adsorption calculation (4.5.1.4.2).

4.5.1.2.3 Calculation. The carbon tetrachloride rate (± 0.1 mg/minute) for each sampling port shall be determined by dividing the weight increase (± 0.1 mg) by the sampling time (± 0.1 minutes). Determine the rate separately for each port. Two or more calculations must agree by ± 0.5 mg/minute in the range of 4.0 to 6.0 mg/minute.

4.5.1.3 Specimen preparation.

4.5.1.3.1 Test specimen. The test specimen shall be a 12.5 ± 0.2 cm diameter circular piece of finished fabric. For each 3 yard sample unit, cut three equally spaced specimens along the left side, three specimens equally spaced along the right side and three specimens equally spaced along the center. Each specimen shall be tested.

4.5.1.3.2 Specimen conditioning. All specimens shall be conditioned in an air circulating oven at 50°C for a minimum of 3 hours prior to adsorption testing. Specimens should be stored in water vapor impermeable containers and exposure to atmospheric moisture should be minimized before adsorption testing.

4.5.1.4 Adsorption test method.

4.5.1.4.1 Procedure. Assemble the complete apparatus as shown on figure 1. Mount a single specimen, face side up in the test cup (figure 2), using three flat rubber rings prepared according to 4.5.1.1.5. The up-side is the inlet side of the carbon tetrachloride vapor mixture flow. First, place one of the 13.80 cm X 11.28 cm rings on the lower cup half. Place a 13.80 cm X 12.70 cm ring on top followed by the test sample so it butts evenly with the narrow ring. Place another 13.80 cm X 12.70 cm ring on top. Place the upper cup half on top. Compress the cups with a "C" type clamp. Place 10 mL of starch/iodide reagent solution mixture (4.5.1.1.12) in each of the bubblers (figure 4) and immerse the bubbler in an ice water bath. A white background behind the bubbler will facilitate detection of the blue color to appear in the bubbler. This establishes the recorded time as the original "break" time. This bubbler shall be replaced with a fresh bubbler which shall be run until the blue color is observed. If more than 2 minutes elapse before the color develops, this bubbler shall be replaced with fresh bubblers until the indicator solution turns blue within a 2 minute period. The true "break time" shall be taken as the total run time prior to the final bubbler. The sample cup shall be purged with air after running each sample to remove any risidual carbon tetrachloride vapor.

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4.5.1.4.2 Calculation. The carbon tetrachloride sample adsorption shall be calculated as follows:

$$mg/cm^2 = \frac{T \times C}{A}$$

Where: T = break time (accurate to 0.1 minute)
C = mg carbon tetrachloride/minute (accurate to 0.1 mg)
A = area of test specimen (100 cm^2)

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, or Commercial. The cloth shall be packed in accordance with with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required in the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1136.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The chemical protective cloth is intended for use in in the manufacture of the Undergarment, Chemical Protective, Two Piece (Undershirt and Drawers).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.3, and 6.3).
d. Length required if other than specified (see 3.5).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to furnished. The contracting officer should also include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Finish treatments. Approval of finish treatments is the responsibility of the U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 and is based on extensive tests including those for toxicity which are not set forth in this specification. Approximately 6 months is required to evaluate and approve new treatments. Only those treatments approved by the U.S. Army Natick Research, Development and Engineering Center will be considered acceptable for the related procurement.

6.5 Toxicity. Due to the toxicity of carbon tetrachloride, the effluent must be vented into a hood. The exit gases of the vacuum pump and manifold must be vented (see 4.5.1.1.1).

6.6 Subject term (key word) listing.

Undergarment

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 8305-A443)

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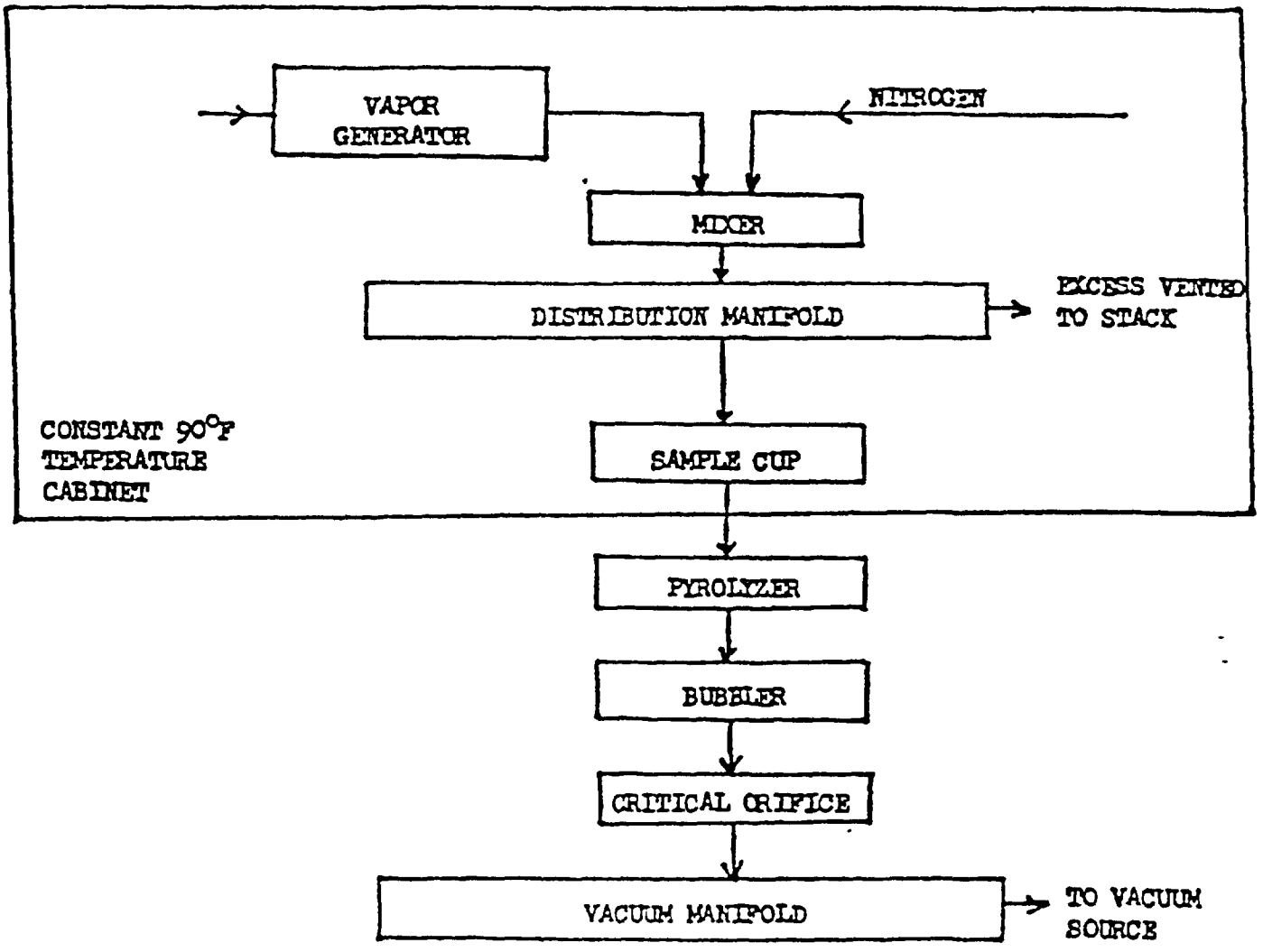
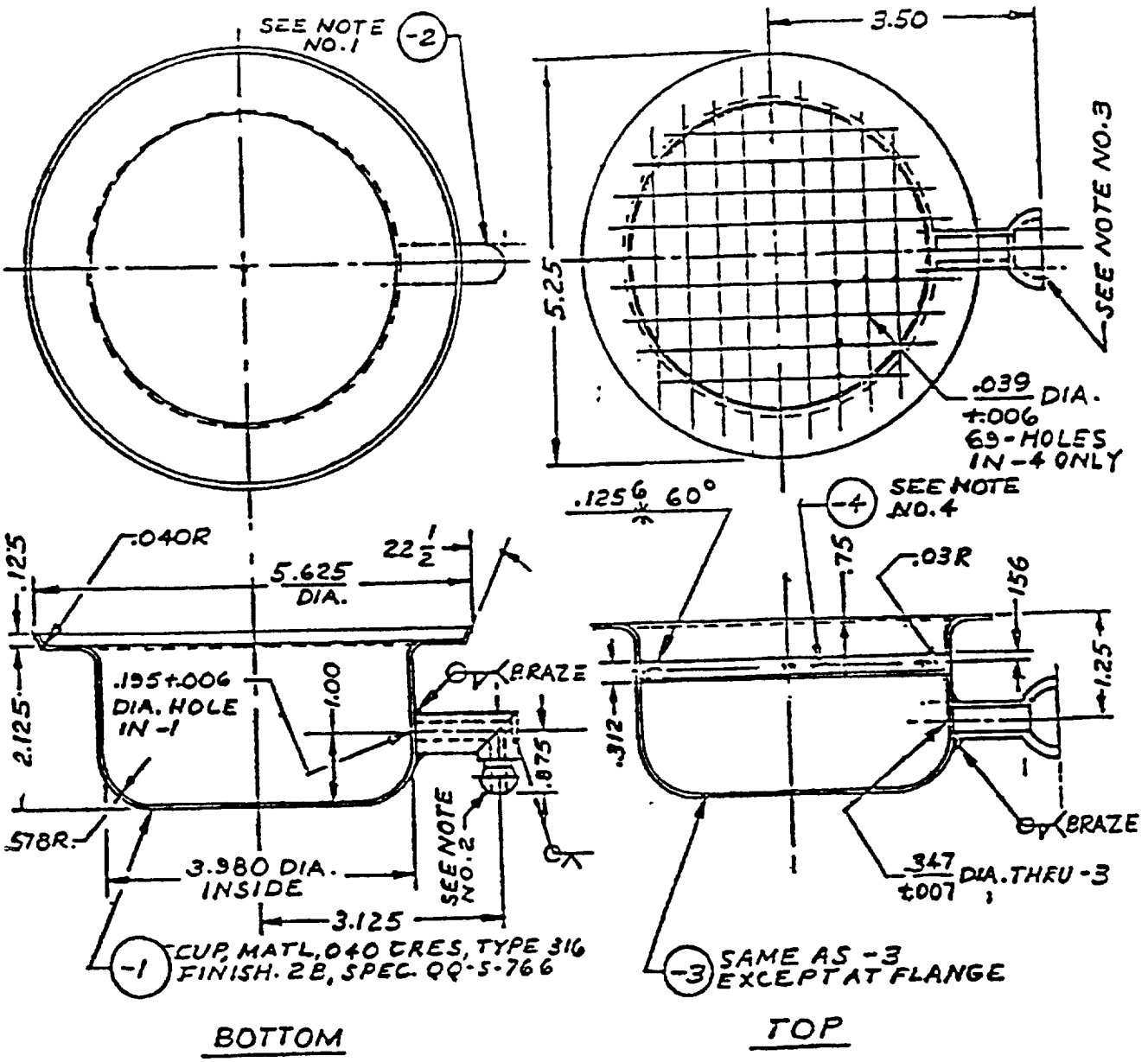


FIGURE 1

FLOW DIAGRAM OF VAPOR PENETRATION APPARATUS

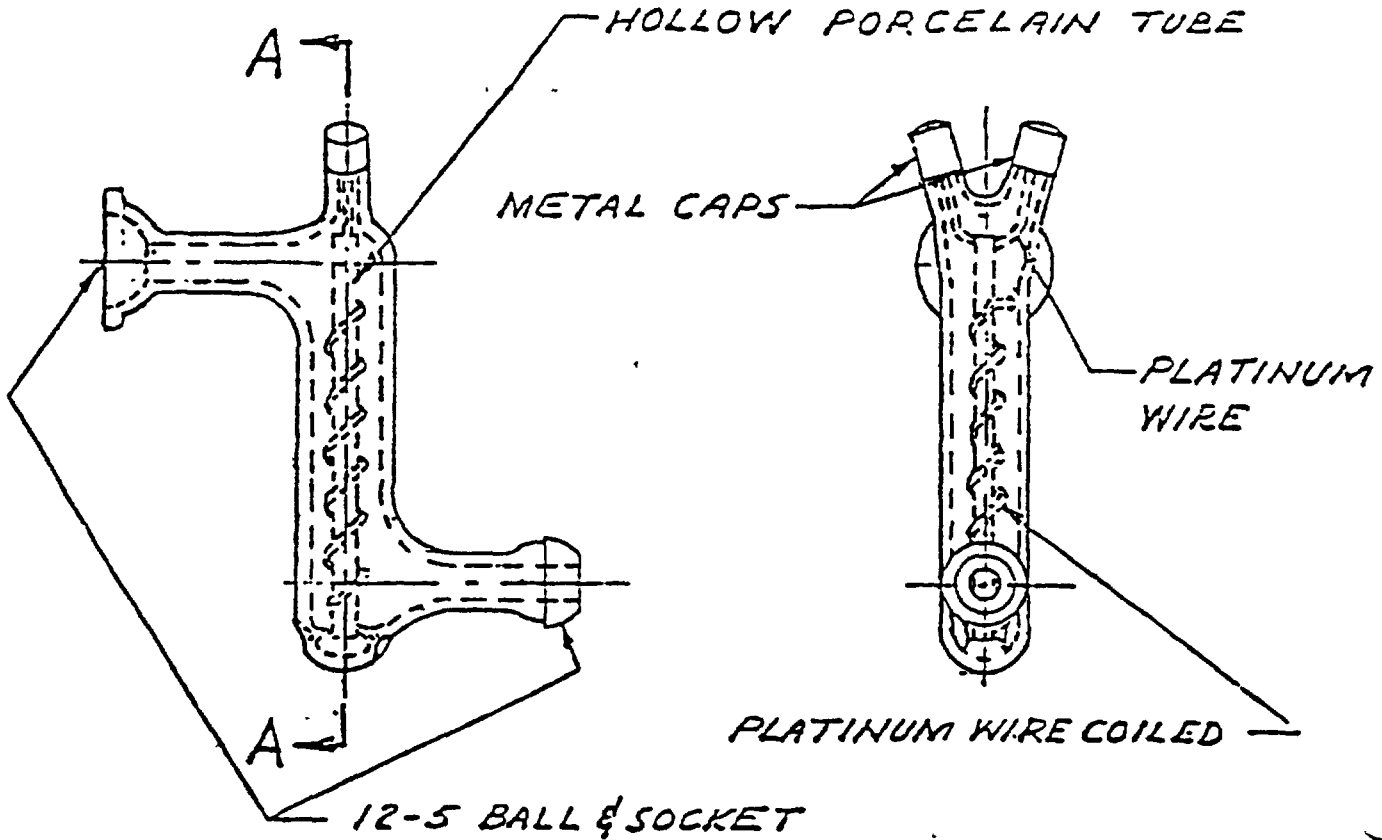
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NOTE:
1. 540-.010 O.D. X .190 ±.006 I.D. CRES TYPE 316, SPEC. QQ-S-763.
2. GROUND JOINT, S, METAL BALL SIZE 12/5, TYPE 316, CRES. KONTES, TECH GLASS CO., VINELAND, N.J.
3. GROUND JOINT S, METAL SOCKET SIZE 18/9 TYPE 316 CRES. KONTES TECH GLASS CO., VINELAND, N.J.
4. MAT'L., .021 THICK, CRES, TYPE 316, FINISH 2B, SPEC QQ-S-766
5. REMOVE ALL BURRS & BREAK SHARP EDGES.

FIGURE 2. FABRIC SAMPLE CUP

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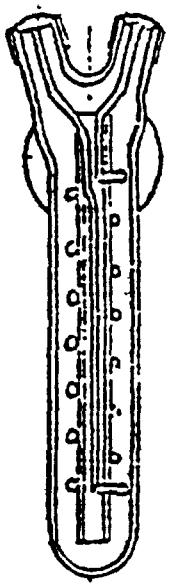


Materials:

(1) 18 gauge platinum wire used in leads.

(2) 30 gauge platinum wire used for heating element. Wire is 36 inches long and is wound around a 0.040 diameter wire to produce a coiled element. The 0.040 diameter wire is removed from the coiled wire. One inch of wire is left uncoiled on each end of the heating element to facilitate fusion to the leads.

(3) The coiled heating element is wrapped around a hollow alundum tube 1/8 inch O.D. and 1/6 inch I.D.



Scale 1/1

SECTION A-A

FIGURE 3
PYROLYZER

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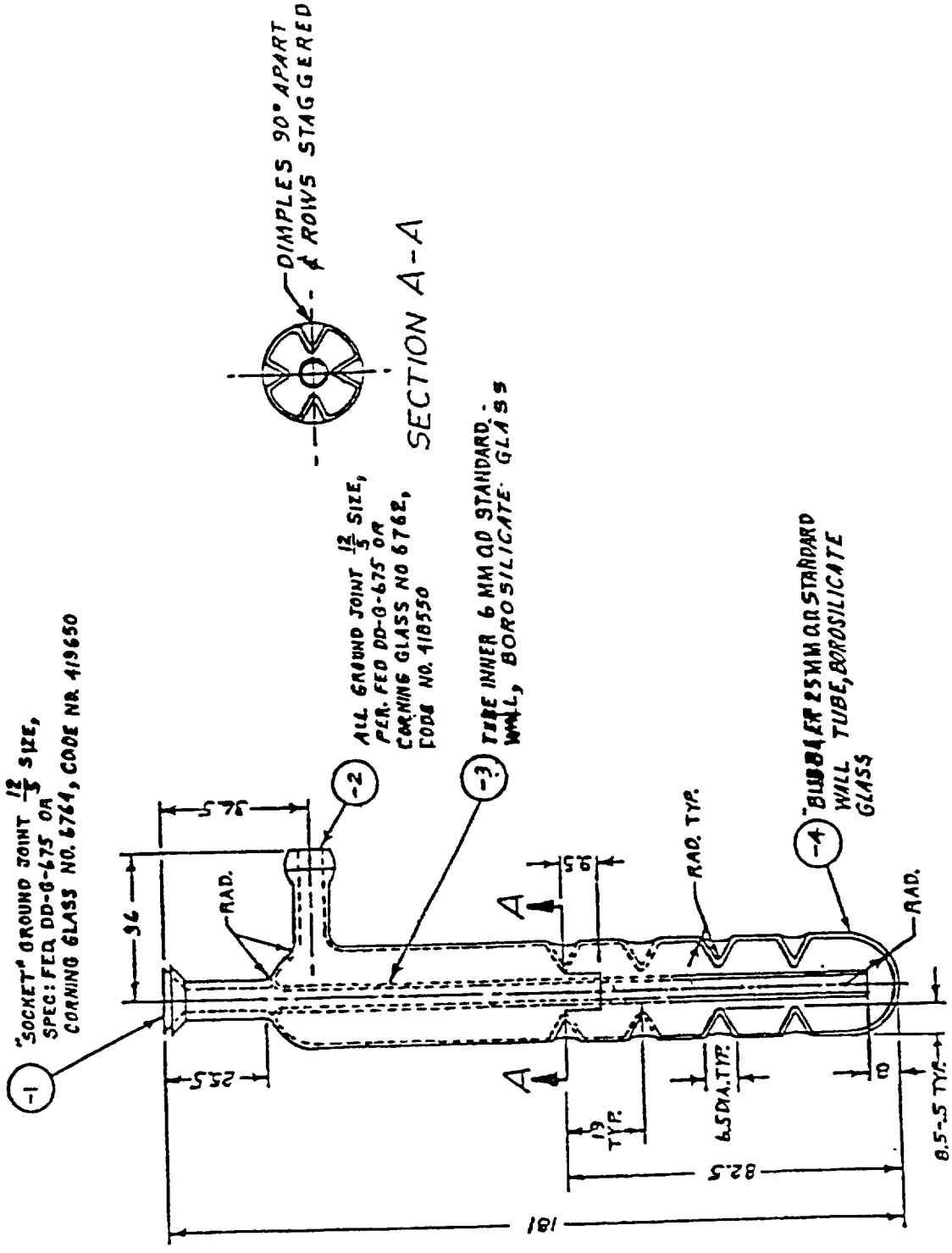


FIGURE 4. BUBBLER

NOTE

1: ALL DIM. ARE IN MILLIMETERS.
TOL.: ± 1 MM

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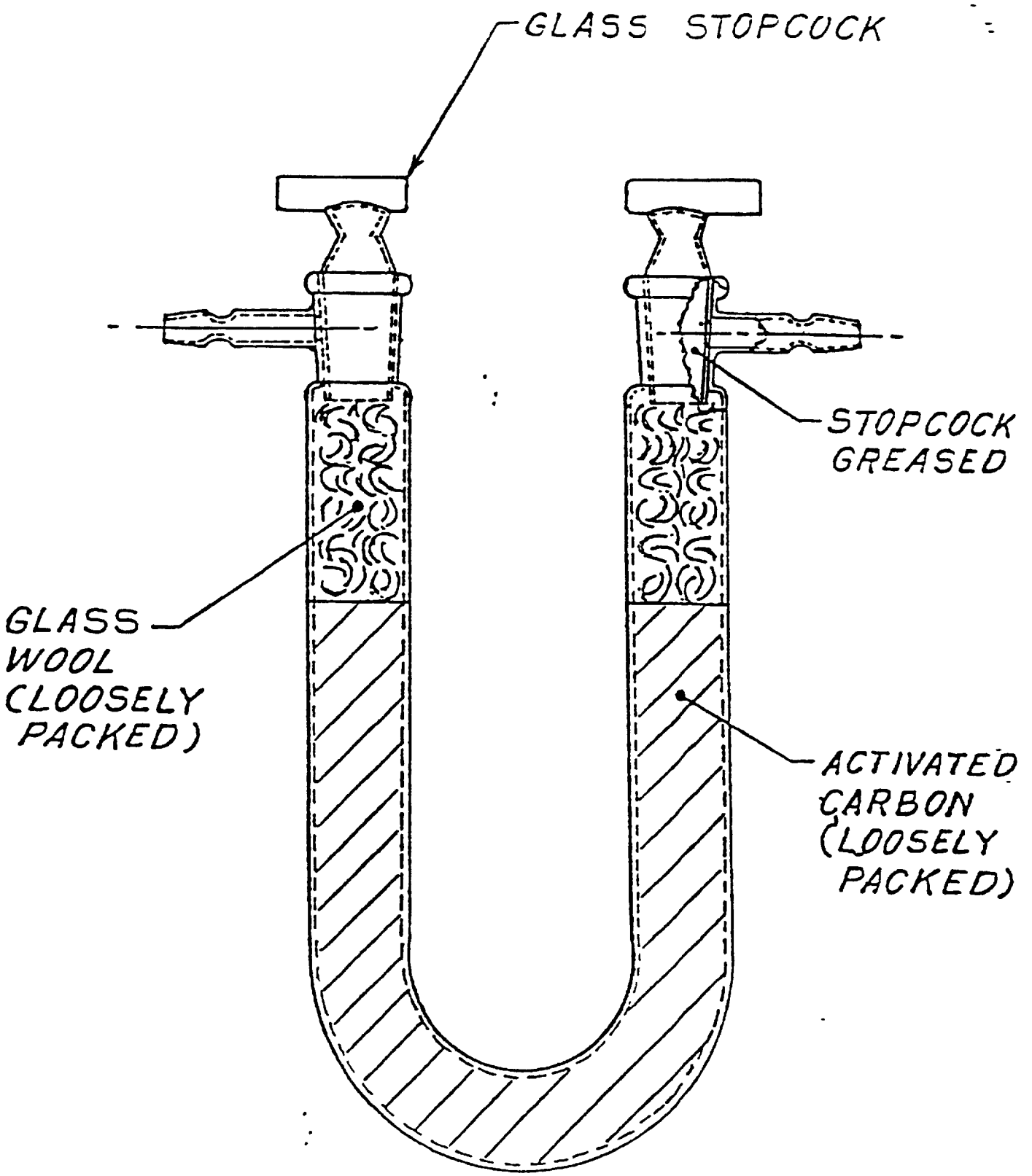


FIGURE 5.

SCHWARTZ TYPE DRYING TUBES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44424(GL)	1991 August 1

3 DOCUMENT TITLE

CLOTH, CHEMICAL PROTECTIVE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial: (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
J.S. Army Natick RD&E Center	508-651-4532	256-4532
c ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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